Problema 6

```
> restart;
> f := x→arctan(x);
```

$$f := x \mapsto \arctan(x) \quad (1)$$

```
> taylor(f(x), x = 0, 7)
```

$$x - \frac{1}{3}x^3 + \frac{1}{5}x^5 + O(x^7) \quad (2)$$

Problema 7

Problema 8. Aproximarea Pade

$$f(x) \approx \frac{p_m(x)}{q_k(x)} = \frac{\sum_{i=0}^{m}(a_i \cdot x^i)}{\sum_{j=0}^{k}(b_j \cdot x^j)} = R_{m,k}(x);$$

$$b_0 = 1;$$

$$c_i = \frac{f^{(i)}(0)}{i!} \text{ si } c_i = 0 \text{ pt } i < 0;$$

Pentru a obtine coeficientii b_i, $i = 1 \ldots k$, *se rezolva sistemul* :

$$\begin{bmatrix} c_m & c_{m-1} & \cdots & c_{m-(k-2)} & c_{m-(k-1)} \\ c_{m+1} & c_m & \cdots & c_{m-(k-3)} & c_{m-(k-2)} \\ \cdots & \cdots & \cdots & \cdots & \cdots \\ c_{m+(k-2)} & c_{m+(k-3)} & \cdots & c_m & c_{m-1} \\ c_{m+(k-1)} & c_{m+(k-2)} & \cdots & c_{m+1} & c_m \end{bmatrix} \begin{bmatrix} b_1 \\ b_2 \\ \cdots \\ b_{k-1} \\ b_k \end{bmatrix} = \begin{bmatrix} -c_{1\,m+1} \\ -c_{m+2} \\ \cdots \\ -c_{m+(k-1)} \\ -c_{m+k} \end{bmatrix}$$

Pentru a obtine coeficientii a_j :

$$a_j = \sum_{l=0}^{j}(c_j - b_l) \text{ pentru } j = 0 \ldots m$$

```
> aproximarea_pade := proc(f, m, k)
    local c, p, C, dreapta, matrice_b, i, j, a, b, l, R;
  description "aproximare pade";
```

```
    # Coeficienții seriei Taylor
    for i from 0 to m + k do
      c[i] := coeftayl(f(x), x = 0, i);
    end do;

  for i from -k to -1 do
    c[i] := 0;
  end do;

    # Matricea Toeplitz
    C := LinearAlgebra:-ToeplitzMatrix([seq(c[m + i], i = -k + 1 .. k - 1)]);
     dreapta := Vector([seq(-c[m + i], i = 1 .. k)]);

    # Verificare singularitate
    if LinearAlgebra:-Determinant(C) = 0 then
      error "Matrix is singular, cannot solve the system.";
    end if;

    # Rezolvare sistem
    matrice_b := LinearAlgebra:-LinearSolve(C, dreapta);

    # Coeficienți numitor
    b[0] := 1;
    for i from 1 to k do
       b[i] := matrice_b[i];
    end do;

    if k ≤ m then
      for i from k + 1 to m do
        b[i] := 0;
      end do;
    end if;

    # Coeficienți numărător
    for j from 0 to m do
      a[j] := sum(c[j - l] * b[l], l = 0 .. j);
     end do;

    # Funcția aproximantă Padé
    R := x → sum(a[ind_i] * x^ind_i, ind_i = 0 .. m) / sum(b[ind_j] * x^ind_j, ind_j = 0 .. k);

    return R(x);
  end proc;

aproximarea_pade := proc(f, m, k)                                    (2.1)
    local c, p, C, dreapta, matrice_b, i, j, a, b, l, R;
```

```
    description "aproximare pade";
    for i from 0 to m + k do c[i] := coeftayl(f(x), x = 0, i) end do;
    for i from -k to -1 do c[i] := 0 end do;
    C := LinearAlgebra:-ToeplitzMatrix([seq(c[m + i], i = -k + 1 ..k - 1)]);
    dreapta := Vector([seq(-c[m + i], i = 1 ..k)]);
    if LinearAlgebra:-Determinant(C) = 0 then
        error "Matrix is singular, cannot solve the system."
    end if;
    matrice_b := LinearAlgebra:-LinearSolve(C, dreapta);
    b[0] := 1;
    for i to k do b[i] := matrice_b[i] end do;
    if k <= m then for i from k + 1 to m do b[i] := 0 end do end if;
    for j from 0 to m do a[j] := sum(c[j - l] * b[l], l = 0 ..j) end do;
    R := x -> sum(a[ind_i] * x^ind_i, ind_i = 0 ..m) / sum(b[ind_j] * x^ind_j, ind_j = 0 ..k);
    return R(x)
end proc
```

```
> f := x -> exp(x);
```

$$f := x \mapsto e^{x} \tag{2.2}$$

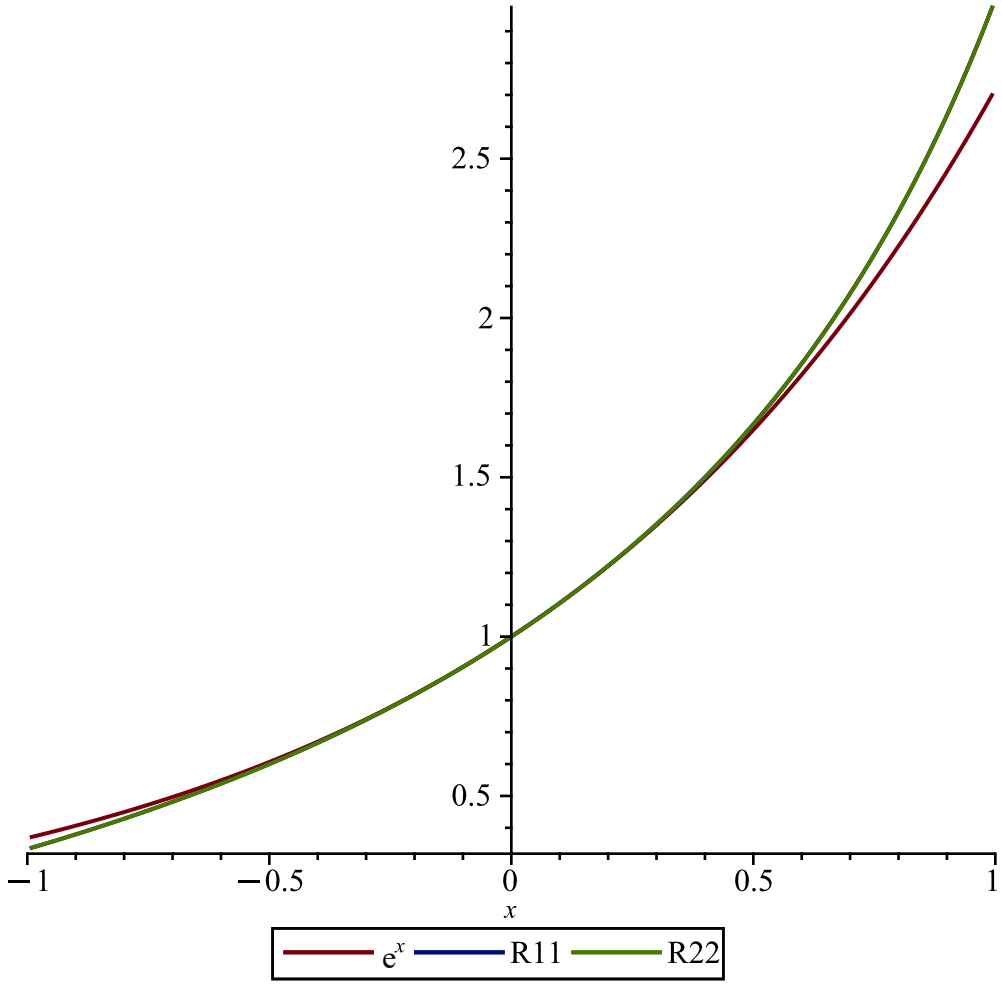
```
> R11f := aproximarea_pade(f, 1, 1);
```

$$R11f := \frac{1 + \frac{x}{2}}{1 - \frac{x}{2}} \tag{2.3}$$

```
> R22f := aproximarea_pade(f, 1, 1);
```

$$R22f := \frac{1 + \frac{x}{2}}{1 - \frac{x}{2}} \tag{2.4}$$

```
> plot([f(x), R11f, R22f], x = -1 ..1, legend = [f(x), "R11", "R22"]);
```



> $g := x \to \ln(1 + x);$

$$g := x \mapsto \ln(x + 1) \tag{2.5}$$

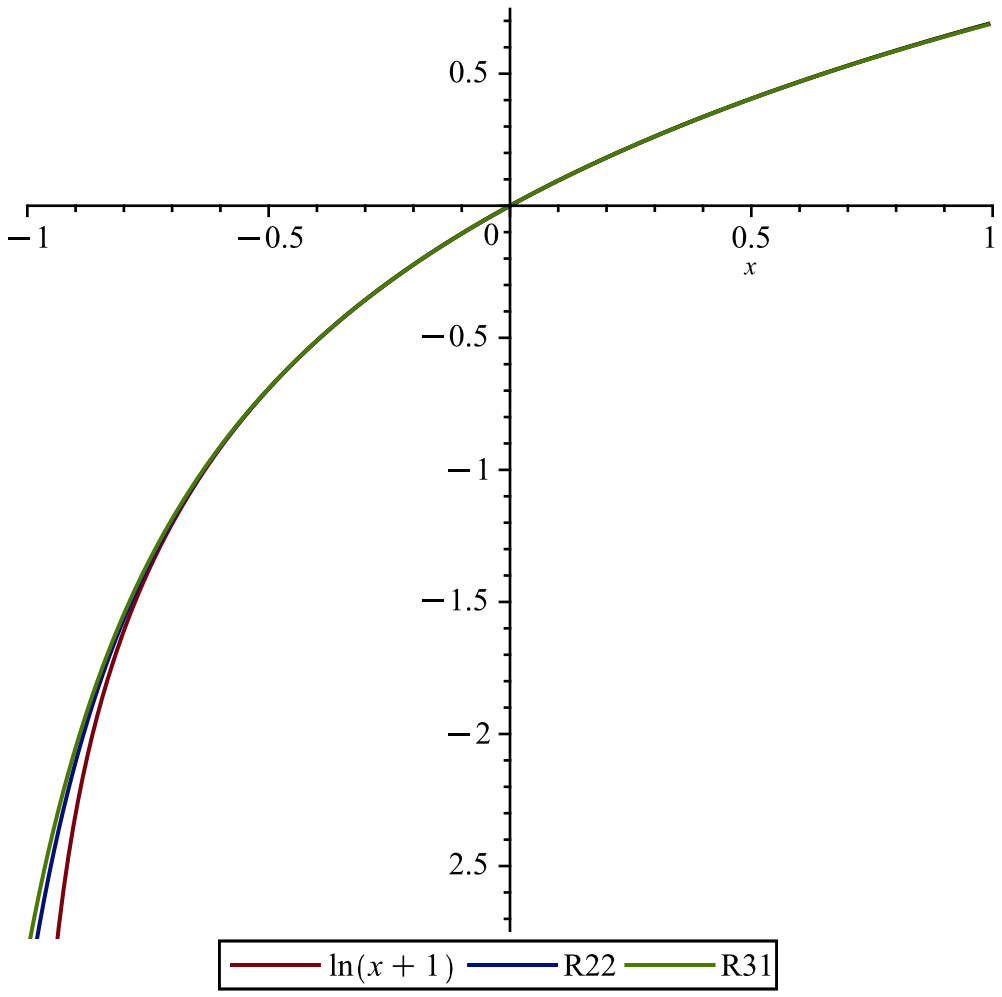
> $R22g := aproximarea_pade(g, 2, 2);$

$$R22g := \frac{x + \frac{1}{2}x^2}{1 + x + \frac{1}{6}x^2} \tag{2.6}$$

> $R31g := aproximarea_pade(g, 3, 1);$

$$R31g := \frac{x + \frac{1}{4}x^2 - \frac{1}{24}x^3}{1 + \frac{3x}{4}} \tag{2.7}$$

> $plot([g(x), R22g, R31g], x = -1..1, legend = [g(x), "R22", "R31"])$



Problema 9

Problema 10

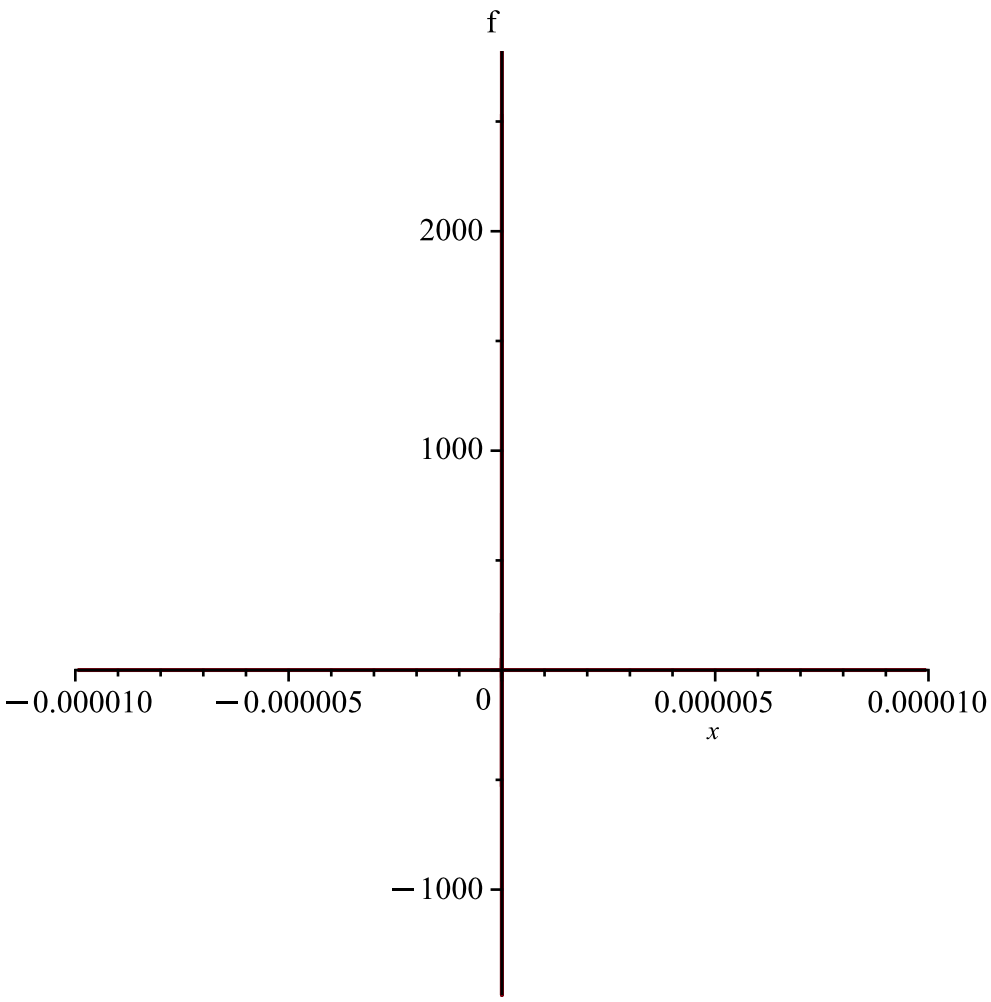
> $f := x \to \frac{(\exp(x) - 1 - x)}{x^2}$

$$f := x \mapsto \frac{e^x - 1 - x}{x^2} \tag{4.1}$$

> $limit(f(x), x = 0)$

$$\frac{1}{2} \tag{4.2}$$

> $plot(f(x), x = -10^{-5}..10^{-5}, title = \text{"f"})$




```
> taylor(f(x), x=0, 5)
```

$$\frac{1}{2} + \frac{1}{6}x + \frac{1}{24}x^2 + O(x^3) \tag{4.3}$$

```
> Tf := x→1/2 + 1/6·x + 1/24·x^2
```

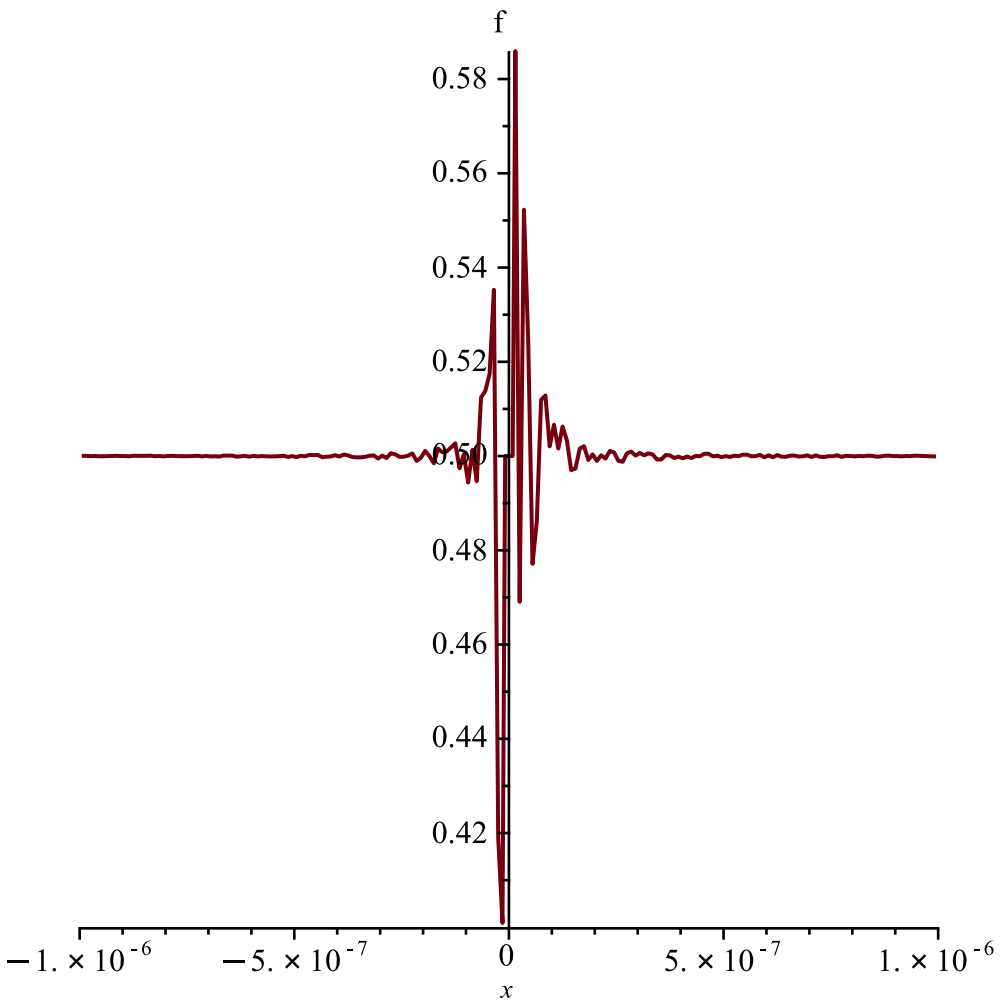
$$Tf := x \mapsto \frac{1}{2} + \frac{1}{6}\cdot x + \frac{1}{24}\cdot x^2 \tag{4.4}$$

```
> epsilon := 10^(-8);
  y := x → piecewise(abs(x) ≤ epsilon, Tf(x), f(x));
```

$$\epsilon := \frac{1}{100000000}$$

$$y := x \mapsto \begin{cases} Tf(x) & |x| \le \epsilon \\ f(x) & otherwise \end{cases} \tag{4.5}$$

```
> plot(y(x), x = -10^(-6) .. 10^(-6), title = "f");
```



Problema 11

$$\left(T_n f\right)(x) = f(a) + \frac{x-a}{1!}\cdot f'(a) + \ldots + \frac{(x-a)^n}{n!}\cdot f^{(n)}(a) = \sum_{i=0}^{n}\frac{(x-a)^i}{i!}\cdot f^{(i)}(a)$$

$$(R_n f)(\mathrm{x}) = \frac{(x-a)^{n+1}\cdot f^{(n+1)}(\xi)}{(n+1)!};\ \xi \in (a,x)$$

```
> formulaLuiTaylorCuRestLagrange := proc(f, a, n)
  local Tnf, Rnf, indice;

      description "Se returneaza aproximarea taylor cu rest in forma Lagrange pentru functia f
      data si rangul n";
  Tnf := x→sum( (x-a)^i/i! · eval(diff(f(x), `$`(x, i)), x = a), i = 0..n );
  Rnf := x→(x-a)^(n+1)·eval(diff(f(x), `$`(x, n+1)), x = ξ)/(n+1)!;
  return Tnf(x) + Rnf(x);
```

```
    end proc;
formulaLuiTaylorCuRestLagrange := proc(f, a, n)
    local Tnf, Rnf, indice;
    description
    "Se returneaza aproximarea taylor cu rest in forma Lagrange pentru functia f data si rangul
    n";
    Tnf := x→sum((x - a)^i * (eval(diff(f(x), x$i), x = a)) / factorial(i), i = 0..n);
    Rnf := x→(x - a)^(n + 1) * (eval(diff(f(x), x$n + 1), x = ξ)) / factorial(n + 1);
    return Tnf(x) + Rnf(x)
end proc
```

(5.1)

```
> f := x→exp(x)
```

$$f := x \mapsto e^{x} \tag{5.2}$$

```
> formulaLuiTaylorCuRestLagrange(f, 0, 5);
```

$$1 + x + \frac{x^2}{2} + \frac{x^3}{6} + \frac{x^4}{24} + \frac{x^5}{120} + \frac{x^6\, e^{\xi}}{720} \tag{5.3}$$

```
>
```